Stage D Direction Detection — Current Best Method

BLE hiker-monitoring project · status as of 2026-08-13

> This is a focused summary of the **currently validated production method** for cross-sensor identity matching and direction detection — not the full investigative history. For the complete series of experiments that led here (RSSI features, order preservation, aggregate flow, transit-time distributions, identity-matching structure, same-MAC evidence, Stage C recall), see `docs/specs/2026-07-30-stage-d-direction-linking-design.md`.

Headline result

HELD-OUT LOSS	DIRECTION SHARE ERROR	CORRELATION	MAE (TOTAL COUNT)
945.0	**0.213**	**0.623**	**17.75**
↓ from 1137.8 pre-follow-up	↓ from 0.263	↑ from 0.512	unchanged

All figures are **leave-one-date-out (LODO)** held-out results across the 8 manual ground-truth dates — every prediction shown was made without that date's own ground truth influencing the parameters used to produce it.

What the method actually is

1. Cross-sensor identity matching

Optimal bipartite assignment (Hungarian algorithm) between Sensor-A and Sensor-B travelers within each time window, on a score combining:

- **Transit-time plausibility** — a bounded Gaussian in the transit gap between a traveler's boundary sessions (peak ≈10s, spread 60s, gated to a max 180s transit / 60s coverage-overlap allowance)
- **Boundary-order agreement** (`order_weight=0.25`) — do the first-seen/last-seen orderings across both sensors agree with the candidate direction
- **Neighborhood-consistency** (`nc_weight=2.0`, 1 neighbor each side) — a bonus for how well a traveler's nearest temporal neighbors' own best candidates agree in relative order. The strongest single addition found: improves held-out loss, share error, and correlation simultaneously at *zero* matched-volume cost, and is the only new scoring term in this project's history chosen identically by every leave-one-date-out fold rather than overfitting one unstable date.
- **Same-MAC bonus** (`mac_bonus=3.0`, binary equality) — a bonus when the two candidate travelers share a raw MAC address across sensors. Rare (~4% of matches) since MAC randomization defeats it for most travelers, but the subset that does share a MAC is measurably cleaner (more

often mutual-best, faster and more physically plausible transit times) and improves the metrics further at negligible cost. Verified non-circular: Stage C's per-sensor clustering never uses raw MAC identity or cross-sensor information, so this is genuinely new evidence.

- **RSSI shape** (`rssi_weight=0.0`) — tested extensively (whole-session trend, local slope, peak timing, full resampled trajectory shape); none of it carries usable directional signal in this dataset, so it's calibrated off.

2. Direction labeling

Simplified to the trivial rule, confirmed equivalent to the full scoring-based choice on **100% of matched pairs** (626/626) before being adopted:

- `last_seen_A < last_seen_B` → **A→B**
- `last_seen_B < last_seen_A` → **B→A**
- exact tie → `first_seen` breaks it; if that's also tied, `ambiguous`

The complexity in the matching score earns its keep entirely in deciding *which pair wins*; it contributes nothing beyond this rule to the direction label of whichever pair does win.

Validated per-date results (held-out)

DATE	GT WINDOW	PRED A→B	PRED B→A	PRED TOTAL	MANUAL A→B	MANUAL B→A	MANUAL TOTAL	PRED A-SHARE	MANUAL A-SHARE	HELD-OUT LOSS
2026-04-11	09:03:54–09:36:57	36	4	40	27	1	28	0.900	0.964	90
2026-05-02	12:36:26–13:15:03	39	36	75	20	36	56	0.520	0.357	361
2026-05-09	11:00:35–11:58:39	186	79	265	181	56	237	0.702	0.764	554
2026-05-16	08:20:51–09:04:17	30	13	43	22	2	24	0.698	0.917	185
2026-05-19	09:27:20–10:43:42	79	35	114	136	2	138	0.693	0.986	4338
2026-05-23	09:04:50–09:57:14	20	28	48	30	13	43	0.417	0.698	325
2026-05-27	10:29:41–10:51:16	16	5	21	17	0	17	0.762	1.000	26
2026-05-31	10:09:30–10:51:04	15	10	25	55	1	56	0.600	0.982	1681

Six of eight dates land within 4–28 people of the true total with directional share trending the right way. The two flagged rows are explained below — they are not a matcher failure.

Progression: what got us here

STAGE	HELD-OUT LOSS	SHARE ERROR	CORR
Timing + boundary-order signal (2026-08-01 state)	1137.8	0.263	0.512
+ neighborhood-consistency	997.4	0.240	0.564
+ same-MAC bonus **(current)**	**945.0**	**0.213**	**0.623**

Two consecutive, independently-validated improvements, each confirmed on real held-out data rather than assumed. RSSI features, hard/soft order-preservation constraints, aggregate flow estimation, candidate pruning, mutual-best-only matching, and anchor-window matching were all tested separately and did *not* beat this configuration — see the design doc for the full negative-result record.

Known limitation — two structurally unreachable dates. 2026-05-19 and 2026-05-31 have a Stage-C traveler count below the manual total no matter how Stage D scores candidates — matching cannot recover volume that was never resolved as distinct travelers upstream. Diagnosed in a dedicated follow-up: 2026-05-31 is a raw BLE-sensing ceiling (Sensor B detected only 40 raw MAC addresses against a manual count of 56 — unfixable at any pipeline stage). 2026-05-19 is genuine Stage C over-merging (crowded periods produce large clusters of temporally-overlapping, similar-RSSI-trend sessions from different people). A candidate fix (`BLOCK_GAP_THRESHOLD_S: 60→30s`) is validated and recommended but **not yet incorporated into this method** — held-out loss 901.0→881.6 if adopted, verified non-pathological across the full 30-day dataset, but it trades some accuracy on 2026-05-09 for improvement on 2026-05-19, so it's staged as a separate decision.

Statistical power. All comparisons in this project are over 8 ground-truth dates. Every metric above should be read as "the best-supported estimate given the data," not as a guarantee — correlation in particular swings substantially between similar-looking configurations at this sample size.

Status: this is the current production configuration. Implemented in `notebooks/04_stage_d_direction_linking.ipynb` , executed against real project data with zero errors, and used to regenerate `DATA/travelers_matched_direction.csv / .parquet` (6,221 matched pairs across all 14 dual-sensor dates). The `direction` column should still be used for aggregate skew rather than individual per-person claims.

Generated from the executed Stage D notebook and the accompanying follow-up investigations logged in `docs/specs/2026-07-30-stage-d-direction-linking-design.md`.